

ADHISH JS

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EDUCATION

National Institute of Technology Puducherry <i>Bachelor of Technology in Electronics and Communication Engineering</i>	Aug 2023 – Present <i>CGPA: 9.75/10</i>
Vijay Vikas International School (CBSE) <i>Senior Secondary (Class XII)</i>	2023 <i>95.2%</i>
Vedanta Academy (CBSE) <i>Secondary (Class X)</i>	2021 <i>92.2%</i>

SKILLS

Programming	Python, C++, Verilog, MATLAB
EDA Tools	Cadence: Virtuoso, Genus, Innovus, Xcelium, JasperGold, OrCAD, Allegro PCB Siemens EDA: Calibre (DRC, LVS, PEX) Synopsys: Sentaurus Workbench (SWB), Sentaurus Structure Editor (SDE), Sentaurus Visual FPGA: Vivado (Xilinx), Libero SoC (Microchip)
Hardware	MCUs: Arduino Uno, Arduino Nano, ESP32, Raspberry Pi Pico SBCs: Raspberry Pi 4 FPGAs: Basys 3, PolarFire Icicle Kit SoC
Simulation	LTspice, QUCS, Multisim

ASIC DESIGN PROJECTS

1D Convolutional Spiking Neural Network (CSNN)	Mar 2026 – May 2026
- Designed a 2 mm × 2 mm ASIC in SCL 180 nm, implementing a 1D CSNN with 16 parallel LIF neurons. - Implemented programmable 1D temporal convolution with on-chip loading support of synaptic weights and thresholds for real-time spike-based processing. - Designed the architecture for edge AI applications such as PCG classification, keyword spotting, event-based gesture recognition, and wearable/IoT sensing through efficient spike-based processing.	
Digital Leaky Integrate-and-Fire (LIF) Neuron Chip	Sep 2025 – Nov 2025
- Taped out a Digital LIF Neuron ASIC on the SCL 180 nm PDK implementing an event-driven spiking neuron architecture for neuromorphic hardware; currently awaiting fabricated silicon for post-tapeout validation. - Implemented digital circuitry for membrane integration, programmable leakage, threshold comparison, spike generation, and refractory reset, providing biologically inspired temporal processing. - Targeted edge AI, event-based vision, robotics, wearable devices, and large-scale SNN accelerators.	

PUBLICATIONS

IEEE ICEE 2025 An Efficient FPGA Implementation of Spiking Neural Networks for N-MNIST Classification
- Designed and implemented a hardware-aware Spiking Neural Network (784–100–10) accelerator on a Xilinx Artix-7 (Basys-3) FPGA - Developed a multiplier-less digital Leaky Integrate-and-Fire (LIF) neuron with time-multiplexed computation, requiring only 4,808 LUTs. - Achieved 96.5% MNIST classification accuracy with approximately 170 mW on-chip power consumption, demonstrating an efficient edge-AI accelerator.

PROJECTS

Low Power Phonocardiogram (PCG) Classification Using 1D CSNN	Jan 2026 – Present
- Designed and trained a 1D CSNN with ≈16K parameters for edge-oriented PCG classification. - Benchmarked five time-frequency feature extraction methods (CQT, Mel Spectrogram, Chirplet, CWT, and MFCC); CQT achieved the highest validation accuracy of 96.8%. - Trained and evaluated the model on PhysioNet/CinC 2016 dataset comprising 3,100+ heart sound recordings.	
FPGA Accelerated PRC-PSS for Enhanced Border Security	Oct 2025 – Dec 2025
- Designed a human-in-the-loop border security system on the Microchip PolarFire Icicle Kit. - Integrated the SCRFD face detection model with the VectorBlox CNN IP on the FPGA fabric, enabling low latency hardware accelerated face detection. - Implemented a multi-threaded C++ application on RISC-V processor for video capture, target tracking, and real-time video streaming.	

Real-Time Animal Detection and Alert System using Raspberry Pi 4

Sep 2024 – Nov 2024

- Developed a real-time wildlife monitoring system on **Raspberry Pi 4** integrating an ultrasonic sensor, USB camera, and a VGG19-based CNN for automated animal detection across 32 species, achieving **92%** classification accuracy.
- Implemented an end-to-end alert pipeline that uploaded detected images to Google Drive and sent SMS notifications via the SMSChef API upon detection.

EXPERIENCE

Indian Institute of Technology Madras (IIT Madras)

May 2026 – July 2026

Summer Research Intern

- Performed transistor and circuit characterization in TSMC 180 nm, and PTM-MG 20 nm FinFET technologies, extracting g_m , g_{ds} , intrinsic gain, f_T , noise, and PVT performance metrics.
- Designed and analyzed CMOS inverters, StrongARM latch comparators, and inverter-based operational amplifiers using Cadence Virtuoso.
- Executed complete custom IC design and verification flow including schematic capture, layout design, DRC, LVS, PEX, and post-layout simulation.
- Designed a 2-layer PCB using OrCAD Capture and Allegro PCB Editor for validation of a digital clock IC.

L&T Precision Engineering & Systems, Coimbatore (Remote)

Mar 2026 – Jun 2026

Space Systems Engineering Intern

- Developed a Python-based Spacecraft Design Suite integrating power budgeting, communication link analysis, hardware power estimation, data budgeting, and memory sizing into a unified desktop application.
- Built modular calculation engines for spacecraft subsystem analysis, enabling rapid system-level trade-off evaluation across power, communication, hardware, data, and storage budgets.

Indian Institute of Science (IISc), Bangalore (Remote)

Dec 2025 – Feb 2026

Research Intern, MSDLab (Dept. of ESE)

- Developed a Python TCL script generator for Sentaurus Visual supporting 3D (cutplane-to-cutline) and 2D TCAD structure analysis, plus transient PLT-based plotting, with both terminal and GUI interfaces.
- Implemented multi-node, multi-TDR, and multi-cutline batch processing with parameter short-code mapping, automating extraction of 1D data curves from complex TCAD simulation files.

India Semiconductor Workforce Development Program (ISWDP)

Feb 2025 – May 2025

- Performed semiconductor process and device simulations using the **Synopsys Sentaurus** TCAD suite for CMOS process and device modeling.
- Received hands-on training in semiconductor device physics, CMOS design, fabrication processes, and VLSI architecture.

ACHIEVEMENTS

- Secured **AIR 835 (Score: 682)** in **GATE 2026 (ECE)**.
- Recognized as a **Top Performer** in Cohort 4 of the ISWDP jointly conducted by IISc, Synopsys, and Samsung Semiconductor India Research.
- Selected as a **Finalist** in the **VLSID Design Contest**.
- School Topper in Grade 12.

LEADERSHIP EXPERIENCE

- **Vice President, ELECSA** – Directed club operations and organized technical workshops and student events.
- **Treasurer, ANTRIX Club** – Managed annual budget and financial planning.
- **Class Representative, NSS** – Organized community initiatives and technical events.
- **Class Committee Member** – Contributed to academic discussions and student affairs.

LANGUAGES

English (Advanced), **Tamil** (Native).